

# DAC Enode Intelligence Watcher - Custom All Time Report

Generated at UTC: 2026-06-29T20:24:53.571881+00:00

Report range: All Time

Project: DAC Enode Intelligence Watcher

This custom report is generated from local watcher JSON outputs.

Important note: this report is observation-based. It does not make official DAC ownership claims and should not be treated as a definitive decentralization measurement.

## 1. Report Scope

| Field                         | Value                            |
|-------------------------------|----------------------------------|
| Range                         | All Time                         |
| Observation count             | 133                              |
| First observed source time    | May 15, 2026 (00:00 CEST)        |
| Last observed source time     | Jun 28, 2026 (19:00 CEST)        |
| Latest watcher checked_at_utc | 2026-06-28T17:12:44.835160+00:00 |
| Latest source generated time  | Sun Jun 28 07:00:02 PM CEST 2026 |

## 2. Enode Movement Summary

| Metric                     | Value |
|----------------------------|-------|
| Minimum enode count        | 2     |
| Maximum enode count        | 16    |
| Average enode count        | 11.46 |
| Total added observations   | 141   |
| Total removed observations | 113   |

| Metric                | Value |
|-----------------------|-------|
| Target ports observed | 28657 |

### 3. Observation Source Coverage

| Phase             | Observations |
|-------------------|--------------|
| automated_watcher | 119          |
| manual_backfill   | 14           |

### 4. Anomaly Summary

| Field                    | Value                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| Selected anomaly signals | 12                                                                                                           |
| Global anomaly summary   | 12 anomaly signals were detected across 133 observations. The highest observed anomaly severity is CRITICAL. |
| Global highest severity  | CRITICAL                                                                                                     |
| Recommended action       | Use these anomaly events as candidates for deeper manual review and future technical reporting.              |

| Severity | Signals in selected range |
|----------|---------------------------|
| HIGH     | 8                         |
| CRITICAL | 2                         |
| MEDIUM   | 2                         |

| Anomaly Type           | Signals in selected range |
|------------------------|---------------------------|
| AGGRESSIVE_ROTATION    | 4                         |
| HIGH_REMOVAL_EVENT     | 3                         |
| LARGE_ENODE_COUNT_DROP | 1                         |
| SHARP_ENODE_COUNT_DROP | 1                         |
| LOW_CONTINUITY_RATIO   | 1                         |

| Anomaly Type                 | Signals in selected range |
|------------------------------|---------------------------|
| WATCHER_HIGH_SEVERITY_SIGNAL | 1                         |
| MODERATE_ROTATION_SPIKE      | 1                         |

## 5. Provider / ASN Concentration Context

| Field                       | Value                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overall concentration label | MODERATE                                                                                                                                                                                                                                                                                                                                   |
| Headline                    | Observed infrastructure shows moderate concentration under the current heuristic model.                                                                                                                                                                                                                                                    |
| Key observation             | Top live ASN is AS51167 with 15 unique IPs (37.5%).                                                                                                                                                                                                                                                                                        |
| Country observation         | Top live ASN country code is DE with 18 unique IPs (45.0%).                                                                                                                                                                                                                                                                                |
| Interpretation              | Top live ASN controls at least 35% of observed unique IPs.                                                                                                                                                                                                                                                                                 |
| Disclaimer                  | Provider concentration and decentralization risk summary is an observation-based heuristic. It is based on currently available watcher data, live ASN enrichment, static provider hints, and DAC Infrastructure Signal labels. It should not be treated as an official DAC classification or as a definitive decentralization measurement. |

| Dimension                 | Top Name                       | Top Count | Top % | Label    |
|---------------------------|--------------------------------|-----------|-------|----------|
| Live ASN                  | AS51167                        | 15        | 37.5  | MODERATE |
| Live Country              | DE                             | 18        | 45.0  | MODERATE |
| DAC Infrastructure Signal | Retained Infrastructure Signal | 14        | 35.0  | MODERATE |

## 6. Observation Timeline

| # | Phase           | Status         | Source Time               | Current | Added | Removed | Unchanged | Port  |
|---|-----------------|----------------|---------------------------|---------|-------|---------|-----------|-------|
| 1 | manual_backfill | manual_initial | May 15, 2026 (00:00 CEST) | 11      | 11    | 0       | 0         | 28657 |
| 2 | manual_backfill | manual_changed | May 16, 2026 (20:00 CEST) | 14      | 4     | 1       | 10        | 28657 |
| 3 | manual_backfill | manual_changed | May 17, 2026 (16:00 CEST) | 13      | 1     | 2       | 12        | 28657 |
| 4 | manual_backfill | manual_changed | May 18, 2026 (08:00 CEST) | 13      | 1     | 1       | 12        | 28657 |

| #  | Phase             | Status           | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|------------------|---------------------------|---------|-------|---------|-----------|-------|
| 5  | manual_backfill   | manual_changed   | May 19, 2026 (16:00 CEST) | 7       | 1     | 7       | 6         | 28657 |
| 6  | manual_backfill   | manual_changed   | May 20, 2026 (20:00 CEST) | 12      | 5     | 0       | 7         | 28657 |
| 7  | manual_backfill   | manual_changed   | May 22, 2026 (00:00 CEST) | 12      | 1     | 1       | 11        | 28657 |
| 8  | manual_backfill   | manual_changed   | May 23, 2026 (12:00 CEST) | 12      | 4     | 4       | 8         | 28657 |
| 9  | manual_backfill   | manual_changed   | May 24, 2026 (04:00 CEST) | 9       | 0     | 3       | 9         | 28657 |
| 10 | manual_backfill   | manual_changed   | May 25, 2026 (04:00 CEST) | 12      | 3     | 0       | 9         | 28657 |
| 11 | manual_backfill   | manual_changed   | May 28, 2026 (08:00 CEST) | 14      | 5     | 3       | 9         | 28657 |
| 12 | manual_backfill   | manual_unchanged | May 29, 2026 (04:00 CEST) | 14      | 0     | 0       | 14        | 28657 |
| 13 | manual_backfill   | manual_changed   | May 30, 2026 (12:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 14 | manual_backfill   | manual_changed   | May 31, 2026 (12:00 CEST) | 15      | 2     | 1       | 13        | 28657 |
| 15 | automated_watcher | initial          | Jun 1, 2026 (04:00 CEST)  | 12      | 12    | 0       | 0         | 28657 |
| 16 | automated_watcher | changed          | Jun 1, 2026 (08:00 CEST)  | 13      | 1     | 0       | 12        | 28657 |
| 17 | automated_watcher | changed          | Jun 1, 2026 (12:00 CEST)  | 9       | 0     | 4       | 9         | 28657 |
| 18 | automated_watcher | changed          | Jun 1, 2026 (22:00 CEST)  | 10      | 1     | 0       | 9         | 28657 |
| 19 | automated_watcher | changed          | Jun 2, 2026 (06:00 CEST)  | 12      | 2     | 0       | 10        | 28657 |
| 20 | automated_watcher | changed          | Jun 2, 2026 (12:00 CEST)  | 12      | 1     | 1       | 11        | 28657 |
| 21 | automated_watcher | changed          | Jun 2, 2026 (22:00 CEST)  | 14      | 2     | 0       | 12        | 28657 |
| 22 | automated_watcher | changed          | Jun 3, 2026 (12:00 CEST)  | 10      | 0     | 4       | 10        | 28657 |
| 23 | automated_watcher | changed          | Jun 3, 2026 (14:00 CEST)  | 8       | 0     | 2       | 8         | 28657 |

| #  | Phase             | Status  | Source Time              | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|--------------------------|---------|-------|---------|-----------|-------|
| 24 | automated_watcher | changed | Jun 3, 2026 (22:00 CEST) | 10      | 2     | 0       | 8         | 28657 |
| 25 | automated_watcher | changed | Jun 4, 2026 (00:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 26 | automated_watcher | changed | Jun 4, 2026 (02:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 27 | automated_watcher | changed | Jun 4, 2026 (12:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 28 | automated_watcher | changed | Jun 4, 2026 (18:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 29 | automated_watcher | changed | Jun 6, 2026 (08:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 30 | automated_watcher | changed | Jun 6, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 31 | automated_watcher | changed | Jun 6, 2026 (14:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 32 | automated_watcher | changed | Jun 6, 2026 (16:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 33 | automated_watcher | changed | Jun 6, 2026 (20:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 34 | automated_watcher | changed | Jun 6, 2026 (22:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 35 | automated_watcher | changed | Jun 7, 2026 (02:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 36 | automated_watcher | changed | Jun 7, 2026 (08:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 37 | automated_watcher | changed | Jun 7, 2026 (10:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 38 | automated_watcher | changed | Jun 7, 2026 (14:00 CEST) | 15      | 1     | 1       | 14        | 28657 |
| 39 | automated_watcher | changed | Jun 7, 2026 (20:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 40 | automated_watcher | changed | Jun 8, 2026 (06:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 41 | automated_watcher | changed | Jun 8, 2026 (20:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 42 | automated_watcher | changed | Jun 8, 2026 (22:00 CEST) | 15      | 1     | 0       | 14        | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 43 | automated_watcher | changed | Jun 9, 2026 (00:00 CEST)  | 14      | 0     | 1       | 14        | 28657 |
| 44 | automated_watcher | changed | Jun 9, 2026 (02:00 CEST)  | 12      | 0     | 2       | 12        | 28657 |
| 45 | automated_watcher | changed | Jun 9, 2026 (06:00 CEST)  | 11      | 0     | 1       | 11        | 28657 |
| 46 | automated_watcher | changed | Jun 9, 2026 (10:00 CEST)  | 10      | 0     | 1       | 10        | 28657 |
| 47 | automated_watcher | changed | Jun 9, 2026 (12:00 CEST)  | 11      | 1     | 0       | 10        | 28657 |
| 48 | automated_watcher | changed | Jun 9, 2026 (16:00 CEST)  | 10      | 0     | 1       | 10        | 28657 |
| 49 | automated_watcher | changed | Jun 9, 2026 (20:00 CEST)  | 11      | 1     | 0       | 10        | 28657 |
| 50 | automated_watcher | changed | Jun 9, 2026 (22:00 CEST)  | 12      | 1     | 0       | 11        | 28657 |
| 51 | automated_watcher | changed | Jun 10, 2026 (06:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 52 | automated_watcher | changed | Jun 10, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 53 | automated_watcher | changed | Jun 10, 2026 (14:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 54 | automated_watcher | changed | Jun 10, 2026 (18:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 55 | automated_watcher | changed | Jun 11, 2026 (00:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 56 | automated_watcher | changed | Jun 11, 2026 (02:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 57 | automated_watcher | changed | Jun 11, 2026 (08:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 58 | automated_watcher | changed | Jun 11, 2026 (12:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 59 | automated_watcher | changed | Jun 11, 2026 (20:00 CEST) | 14      | 1     | 0       | 13        | 28657 |
| 60 | automated_watcher | changed | Jun 11, 2026 (22:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 61 | automated_watcher | changed | Jun 12, 2026 (16:00 CEST) | 14      | 0     | 1       | 14        | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 62 | automated_watcher | changed | Jun 12, 2026 (20:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 63 | automated_watcher | changed | Jun 12, 2026 (22:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 64 | automated_watcher | changed | Jun 13, 2026 (00:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 65 | automated_watcher | changed | Jun 13, 2026 (04:00 CEST) | 16      | 1     | 0       | 15        | 28657 |
| 66 | automated_watcher | changed | Jun 13, 2026 (08:00 CEST) | 15      | 0     | 1       | 15        | 28657 |
| 67 | automated_watcher | changed | Jun 13, 2026 (22:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 68 | automated_watcher | changed | Jun 14, 2026 (02:00 CEST) | 15      | 1     | 0       | 14        | 28657 |
| 69 | automated_watcher | changed | Jun 14, 2026 (12:00 CEST) | 14      | 0     | 1       | 14        | 28657 |
| 70 | automated_watcher | changed | Jun 14, 2026 (14:00 CEST) | 16      | 2     | 0       | 14        | 28657 |
| 71 | automated_watcher | changed | Jun 14, 2026 (18:00 CEST) | 14      | 0     | 2       | 14        | 28657 |
| 72 | automated_watcher | changed | Jun 14, 2026 (20:00 CEST) | 14      | 1     | 1       | 13        | 28657 |
| 73 | automated_watcher | changed | Jun 15, 2026 (08:00 CEST) | 13      | 0     | 1       | 13        | 28657 |
| 74 | automated_watcher | changed | Jun 15, 2026 (20:00 CEST) | 11      | 0     | 2       | 11        | 28657 |
| 75 | automated_watcher | changed | Jun 16, 2026 (04:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 76 | automated_watcher | changed | Jun 16, 2026 (14:00 CEST) | 11      | 2     | 1       | 9         | 28657 |
| 77 | automated_watcher | changed | Jun 16, 2026 (18:00 CEST) | 8       | 0     | 3       | 8         | 28657 |
| 78 | automated_watcher | changed | Jun 16, 2026 (22:00 CEST) | 9       | 1     | 0       | 8         | 28657 |
| 79 | automated_watcher | changed | Jun 17, 2026 (06:00 CEST) | 10      | 1     | 0       | 9         | 28657 |
| 80 | automated_watcher | changed | Jun 17, 2026 (12:00 CEST) | 2       | 0     | 8       | 2         | 28657 |

| #  | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 81 | automated_watcher | changed | Jun 17, 2026 (16:00 CEST) | 6       | 5     | 1       | 1         | 28657 |
| 82 | automated_watcher | changed | Jun 17, 2026 (18:00 CEST) | 7       | 1     | 0       | 6         | 28657 |
| 83 | automated_watcher | changed | Jun 17, 2026 (20:00 CEST) | 7       | 1     | 1       | 6         | 28657 |
| 84 | automated_watcher | changed | Jun 18, 2026 (00:00 CEST) | 8       | 1     | 0       | 7         | 28657 |
| 85 | automated_watcher | changed | Jun 18, 2026 (08:00 CEST) | 7       | 0     | 1       | 7         | 28657 |
| 86 | automated_watcher | changed | Jun 18, 2026 (13:00 CEST) | 10      | 3     | 0       | 7         | 28657 |
| 87 | automated_watcher | changed | Jun 18, 2026 (16:00 CEST) | 9       | 1     | 2       | 8         | 28657 |
| 88 | automated_watcher | changed | Jun 18, 2026 (19:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 89 | automated_watcher | changed | Jun 19, 2026 (00:00 CEST) | 9       | 1     | 0       | 8         | 28657 |
| 90 | automated_watcher | changed | Jun 19, 2026 (07:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 91 | automated_watcher | changed | Jun 19, 2026 (11:00 CEST) | 6       | 1     | 3       | 5         | 28657 |
| 92 | automated_watcher | changed | Jun 19, 2026 (15:00 CEST) | 9       | 3     | 0       | 6         | 28657 |
| 93 | automated_watcher | changed | Jun 19, 2026 (18:00 CEST) | 9       | 1     | 1       | 8         | 28657 |
| 94 | automated_watcher | changed | Jun 19, 2026 (20:00 CEST) | 8       | 0     | 1       | 8         | 28657 |
| 95 | automated_watcher | changed | Jun 19, 2026 (22:00 CEST) | 8       | 1     | 1       | 7         | 28657 |
| 96 | automated_watcher | changed | Jun 20, 2026 (00:00 CEST) | 8       | 1     | 1       | 7         | 28657 |
| 97 | automated_watcher | changed | Jun 20, 2026 (05:00 CEST) | 10      | 2     | 0       | 8         | 28657 |
| 98 | automated_watcher | changed | Jun 20, 2026 (16:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 99 | automated_watcher | changed | Jun 20, 2026 (21:00 CEST) | 11      | 1     | 1       | 10        | 28657 |



| #   | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 100 | automated_watcher | changed | Jun 21, 2026 (00:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 101 | automated_watcher | changed | Jun 21, 2026 (01:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 102 | automated_watcher | changed | Jun 21, 2026 (09:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 103 | automated_watcher | changed | Jun 21, 2026 (15:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 104 | automated_watcher | changed | Jun 21, 2026 (22:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 105 | automated_watcher | changed | Jun 21, 2026 (23:00 CEST) | 10      | 1     | 1       | 9         | 28657 |
| 106 | automated_watcher | changed | Jun 22, 2026 (09:00 CEST) | 8       | 0     | 2       | 8         | 28657 |
| 107 | automated_watcher | changed | Jun 22, 2026 (15:00 CEST) | 6       | 0     | 2       | 6         | 28657 |
| 108 | automated_watcher | changed | Jun 22, 2026 (19:00 CEST) | 9       | 3     | 0       | 6         | 28657 |
| 109 | automated_watcher | changed | Jun 22, 2026 (22:00 CEST) | 10      | 2     | 1       | 8         | 28657 |
| 110 | automated_watcher | changed | Jun 23, 2026 (00:00 CEST) | 9       | 0     | 1       | 9         | 28657 |
| 111 | automated_watcher | changed | Jun 23, 2026 (02:00 CEST) | 10      | 1     | 0       | 9         | 28657 |
| 112 | automated_watcher | changed | Jun 23, 2026 (06:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 113 | automated_watcher | changed | Jun 23, 2026 (10:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 114 | automated_watcher | changed | Jun 23, 2026 (16:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 115 | automated_watcher | changed | Jun 23, 2026 (18:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 116 | automated_watcher | changed | Jun 24, 2026 (01:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 117 | automated_watcher | changed | Jun 24, 2026 (03:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 118 | automated_watcher | changed | Jun 24, 2026 (07:00 CEST) | 13      | 1     | 0       | 12        | 28657 |

| #   | Phase             | Status  | Source Time               | Current | Added | Removed | Unchanged | Port  |
|-----|-------------------|---------|---------------------------|---------|-------|---------|-----------|-------|
| 119 | automated_watcher | changed | Jun 24, 2026 (14:00 CEST) | 12      | 0     | 1       | 12        | 28657 |
| 120 | automated_watcher | changed | Jun 24, 2026 (22:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 121 | automated_watcher | changed | Jun 25, 2026 (00:00 CEST) | 10      | 0     | 1       | 10        | 28657 |
| 122 | automated_watcher | changed | Jun 25, 2026 (21:00 CEST) | 7       | 0     | 3       | 7         | 28657 |
| 123 | automated_watcher | changed | Jun 26, 2026 (02:00 CEST) | 10      | 3     | 0       | 7         | 28657 |
| 124 | automated_watcher | changed | Jun 26, 2026 (10:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 125 | automated_watcher | changed | Jun 26, 2026 (21:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 126 | automated_watcher | changed | Jun 27, 2026 (05:00 CEST) | 10      | 0     | 2       | 10        | 28657 |
| 127 | automated_watcher | changed | Jun 27, 2026 (08:00 CEST) | 11      | 1     | 0       | 10        | 28657 |
| 128 | automated_watcher | changed | Jun 27, 2026 (11:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 129 | automated_watcher | changed | Jun 27, 2026 (16:00 CEST) | 13      | 1     | 0       | 12        | 28657 |
| 130 | automated_watcher | changed | Jun 28, 2026 (10:00 CEST) | 12      | 0     | 1       | 12        | 28657 |
| 131 | automated_watcher | changed | Jun 28, 2026 (12:00 CEST) | 11      | 0     | 1       | 11        | 28657 |
| 132 | automated_watcher | changed | Jun 28, 2026 (14:00 CEST) | 12      | 1     | 0       | 11        | 28657 |
| 133 | automated_watcher | changed | Jun 28, 2026 (19:00 CEST) | 13      | 1     | 0       | 12        | 28657 |

## 7. Report-Use Notes

- Use the 7D report for short-range rotation and recent watcher movement.
- Use the 30D report for broader trend context once the 15-minute watcher has accumulated enough observations.
- Use the All Time report for full testnet observation history preserved by this watcher.

- For publication-quality interpretation, compare this Markdown output with dashboard charts, raw JSON outputs, and selected snapshots.